

How Much Do Time-Series Anomaly Detection Metrics Actually Disagree?

Null Models and Cluster-Aware Inference for Rank-Flip Statistics

Anonymous authors
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Abstract

Reports that point-level and segment-level metrics rank time-series anomaly detectors differently have become a standard argument for evaluation reform. These reports take the form of a rank-flip rate: the fraction of model pairs that two metrics order differently. We audit that statistic on a 180-series, 25-model recomputation of TSB-AD-M spanning 17 source collections, and on a 6-dataset, seven-detector grid. Our main finding is that evaluation series are treated as independent when they are not, and that this invalidates the structural explanations built on top of the statistic. The covariate most often invoked to explain disagreement, the short-anomaly ratio α , is constant within 16 of our 17 collections: it is a collection label rather than a series-level variable. Its rank correlation with the flip rate falls from 0.3241 at the series level to 0.0563 ($p = 0.827$) with collections as the unit, and to 0.0898 when one collection is removed. Interval estimates move the same way, from $[0.2974, 0.3319]$ resampling series to $[0.2701, 0.3695]$ clustering over collections; on the six-dataset grid the clustered interval spans $[0.0667, 0.4167]$, which cannot support any claim about magnitude. Two further omissions point the same way. Raw flip percentages are reported without stating the chance level, which is 0.5 and not 0: our observed 0.3145 against a permutation null of 0.5002 means the metrics agree on 68.6% of pairs. And flips are pooled across margins, though they run from 0.4743 for pairs within 0.01 AUC-ROC to 0.2015 for pairs separated by at least 0.20—the residual disagreement is real but far smaller than the pooled figure suggests. We apply the audit to a composite metric we previously proposed ourselves and show it is uninformative by construction at both ends of the regime it was built to span. Mean segment duration and segment count do survive clustering. We give a reporting protocol and release code that regenerates every number here from committed artifacts.

1 Introduction

Time-series anomaly detection (TSAD) evaluation has been under sustained criticism for a decade. Point-adjustment inflates scores to the point where random detectors appear competitive (Kim et al., 2022); widely used benchmarks contain mislabeled, trivially separable, or unrepresentative anomalies (Wu and Keogh, 2023; Liu and Paparrizos, 2024); and a series of replacement metrics has been proposed to repair specific failures of point-wise scoring (Tatbul et al., 2018; Huet et al., 2022; Paparrizos et al., 2022; Ghorbani et al., 2024).

A particular kind of evidence recurs throughout this literature. One computes two metrics over the same detector–dataset grid, counts the model pairs that the two metrics order differently, and reports the fraction. We call this a rank-flip rate. It is rhetorically powerful: a statement that a third of published rankings would reverse under a defensible alternative metric implies that leaderboard positions carry little information. We used exactly this argument in earlier work.

¹Large language models were used as assistive tools in the statistical re-analysis and in drafting this manuscript. All reported numbers are produced by the released scripts and were verified against the committed artifacts by the author, who takes full responsibility for the content.

This paper asks what a rank-flip rate actually measures. Our finding is that the statistic as reported is missing three ingredients, that each omission pushes the interpretation in the same direction—toward overstating instability—and that supplying them changes several conclusions in this literature, including our own.

A stated chance level. A flip rate is a fraction between 0 and 1, and it is discussed as though 0 were the reference point, so that any positive value reads as evidence of disagreement. The correct reference is the value the statistic takes when the two metrics share no information about model quality, which is 0.5. We are careful about the scope of this claim. The field does baseline against chance elsewhere: Yang et al. (2025) evaluate metric values under random detection, and Lyu (2026) reports inter-metric Kendall’s τ , which is itself chance-corrected. What is missing is narrower and specific: raw flip percentages are reported and interpreted without the chance level being stated alongside them. The move of supplying a permutation null for pairwise reversal rates has also been made independently and concurrently outside our field—Kendiukhov (2026) does exactly this for gene regulatory network benchmarking, obtaining a null of 0.500 against an observed 0.163—which we take as corroboration rather than as a contribution of ours.

A margin treatment. Whether two metrics order a pair of models differently is only meaningful if the models are actually distinguishable. Pairs separated by a negligible margin flip at close to the chance rate by construction, and they make up a substantial share of every grid. Stratifying by margin separates the part of the disagreement that is an artifact of pooling near-ties from the part that is not.

A cluster unit. Evaluation series are routinely treated as independent replicates. They are not: benchmark suites are assembled from source collections, and series within a collection share a sensor layout, a labeling convention, and an anomaly-generating process. This matters most for exactly the covariates used to explain disagreement, because those covariates tend to be properties of the collection rather than of the series. When that is the case, a structural explanation supported at $n = 180$ can be supported by, in effect, a handful of independent observations.

Contributions. Our primary contribution is the third item below; the first two are corrections that are individually straightforward and, in the case of the null, partly concurrent with work in another field.

1. We supply all three ingredients on two grids: a 180-series, 25-model, 4,498-row recomputation of TSB-AD-M spanning 17 source collections, and a 6-dataset, seven-detector study on TAB-style benchmarks (Section 2).
2. We show that reported flip rates describe agreement once the chance level is stated, but that the residual disagreement is not a margin artifact (Sections 3–4).
3. We show that structure-conditioned explanations of disagreement are confounded with collection identity, and that the standard thresholded short-anomaly ratio is not identifiable at the series level at all. We are not aware of a precedent for this analysis of rank-flip statistics, in TSAD or elsewhere (Section 5).
4. We give a worked failure case: a metric defined as an interpolation between two others is uninformative at both endpoints of the very regime it was introduced to span. The metric is one we proposed ourselves, and the analysis we published in support of it cannot come out any other way (Section 6).
5. We identify two continuous covariates that do survive clustering, and give a reporting protocol (Sections 7–8).

We are not arguing that TSAD evaluation is sound. Two of the criticisms above survive our audit intact, and a third—that benchmark anomalies are structurally unlike the point-wise events that point metrics score—we confirm as a factual matter. We are arguing that the specific statistic most often used to quantify the problem does not support the weight placed on it, and that this is fixable.

Table 1: Rank-flip rates against a random-ranking null. Reported rates are routinely discussed against an implicit reference of zero; the informative reference is the null column. All rates are AUC-ROC versus Aff-F1.

Grid	Models	Flips / pairs	Rate	Null	Agreement
TSB-AD-M	25	—	0.3145	0.5002	68.6%
TAB	5 deep	14/60	0.2333	0.5004	76.7%
TAB	7 (with classical)	44/126	0.3492	0.5021	65.1%

2 Setup

Grids. We use two evaluation grids. The first is a 6-dataset study on TAB-style benchmarks (Qiu et al., 2025): SWaT (Mathur and Tippenhauer, 2016; Goh et al., 2017), MSL and SMAP (Hundman et al., 2018), WADI (Ahmed et al., 2017), PSM, and SMD (Su et al., 2019), evaluated with five deep detectors—Anomaly Transformer (Xu et al., 2022), TranAD (Tuli et al., 2022), CATCH (Wu et al., 2025), DADA (Shentu et al., 2025), and MOMENT (Goswami et al., 2024)—plus two classical baselines, Isolation Forest and LOF. The second is a replication on the TSB-AD-M evaluation list (Liu and Paparrizos, 2024), covering 180 multivariate series and 25 scored models for 4,498 model-series metric rows, drawn from 17 source collections.

Metrics. We report AUC-ROC and segment-level affiliation F1 (Huet et al., 2022) computed with prts at cardinality one and flat bias, with thresholds set from the contamination rate of the test partition following TAB’s protocol (Qiu et al., 2025). Contamination-rate thresholding is itself an assumption with known sensitivity (Masakuna et al., 2024; Perini et al., 2023); our analysis conditions on it throughout and we do not claim results are invariant to it.

Rank-flip rate. For a group g (a dataset, or a series) with model scores a_i under a primary metric and b_i under a secondary metric, we count unordered pairs (i, j) with $a_i \neq a_j$ and $b_i \neq b_j$, and call a pair flipped when $\text{sign}(a_i - a_j) \neq \text{sign}(b_i - b_j)$. The flip rate pools flipped pairs over comparable pairs. This matches the definition used in the sources we are auditing, including our own.

Anomaly-duration taxonomy. Anomaly segments are contiguous runs of positive labels in the test partition, bucketed as Short (≤ 5 steps), Medium (6–50), or Long (> 50). The structure weight in common use is $\alpha = 1 - N_{\text{short}}/N_{\text{total}}$. On the 6-dataset grid, only PSM ($\alpha = 0.4861$) and SMD ($\alpha = 0.6147$) have $\alpha < 1$; SWaT, MSL, SMAP, and WADI have no Short segments at all. That last fact is a property of the labels and is not disputed here.

Reproducibility. Every number in this paper is emitted as a $\text{\LaTeX}$ macro by a script that reads the committed JSON artifacts, so the manuscript cannot drift from the analysis. Rerunning three scripts regenerates all artifacts and all numbers.

3 Against a null, the metrics mostly agree

We build the null by permuting the secondary metric across models within each group, which destroys any association with the primary metric while preserving both marginal distributions and all group sizes. Under independent rankings the expected flip rate is 0.5.

Table 1 gives the result. On TSB-AD-M the observed AUC-ROC-versus-Aff-F1 rate is 0.3145 against a null of 0.5002 (sd 0.0056, 200 permutations). On the 6-dataset grid the deep-model rate is $14/60 = 0.2333$ against 0.5004, and the seven-model rate is $44/126 = 0.3492$ against 0.5021. Every observed rate is far below its null.

Table 2: TSB-AD-M flip rate by AUC-ROC margin. Disagreement falls steeply with margin but does not vanish.

$ \Delta\text{AUC-ROC} $	Pairs	Share	Flip rate
< 0.01	4,700	9.8%	0.4743
0.01–0.02	3,139	6.5%	0.4839
0.02–0.05	6,513	13.5%	0.4230
0.05–0.10	7,757	16.1%	0.3616
0.10–0.20	10,026	20.8%	0.2603
≥ 0.20	16,045	33.3%	0.2015

The consequence is a reframing rather than a refutation. “65.1% of pairwise rankings are preserved across a point metric and a segment metric” and “0.3492 of pairwise rankings flip” are the same measurement. The first is the honest summary; the second, which is how such numbers are presented, invites the reader to supply a reference point of zero.

We claim no novelty for this step. Kendiukhov (2026) applies the same permutation construction to ranking reversals in gene regulatory network benchmarking and reaches the analogous conclusion, that a headline reversal rate of 0.163 against a null of 0.500 indicates substantial shared ranking structure. Within TSAD, chance-corrected rank agreement is also already available in the form of Kendall’s τ (Lyu, 2026). The point is that the raw percentage is what circulates, and it circulates without its baseline.

4 The disagreement is not a margin artifact

Two models separated by 0.001 AUC-ROC are not meaningfully ordered by AUC-ROC, so a metric that orders them differently has not reversed a finding. Because such pairs are numerous, they can carry a flip rate on their own.

Table 2 stratifies the TSB-AD-M pairs by $|\Delta\text{AUC-ROC}|$. The pattern is monotone and steep: pairs separated by less than 0.01 flip at 0.4743, essentially the chance rate, while pairs separated by at least 0.20 flip at 0.2015. Near-ties therefore do inflate the pooled statistic.

They do not explain it. Excluding every pair within 0.05 AUC-ROC retains 70.2% of pairs and still gives 0.2556. Among pairs separated by at least 0.20 AUC-ROC—a margin no one would call ambiguous—about one in five still reverses under Aff-F1. This is the part of the phenomenon that survives, and in our view it is the part worth reporting.

A related concern is that flips are produced by detectors that do not discriminate at all. On the 6-dataset grid we drop every model–dataset row with $|\text{AUC-ROC} - 0.5| < \delta$ and recompute. The rate is 0.3492 at $\delta = 0$, then 0.3200, 0.3774, and $13/35 = 0.3714$ at $\delta = 0.02, 0.05, 0.10$. Removing uninformative detectors does not remove the disagreement.

Two cautions about this check. The surviving sample is small—35 pairs at $\delta = 0.10$ —and the clustered interval is correspondingly wide, $[0.1765, 0.6155]$. And excluding by measured discriminative power is not the same as excluding by model family: Isolation Forest is the strongest detector by AUC-ROC on four of the six datasets, while Anomaly Transformer sits within 0.03 of chance on all six, so the $\delta = 0.10$ exclusion drops 16 deep rows against 5 classical ones. Pairs that mix a deep and a classical detector do flip more often ($27/60 = 0.4500$, versus $14/60 = 0.2333$ for deep–deep pairs), but this traces to LOF specifically rather than to classical methods as a class.

5 Evaluation series are not independent

TSB-AD-M draws its 180 series from 17 source collections. Pooled flip rates differ sharply across them, from 0.1311 on TAO to 0.5876 on GHL. Treating series as independent replicates therefore overstates precision, and—more seriously—creates a confound for any covariate that is a property of the collection.

Table 3: Within-collection variation of structural covariates on TSB-AD-M. A covariate that is constant within collections cannot be identified at the series level.

Covariate	Constant within (of 17 collections)	Constant within large (of 8 with ≥ 8 series)
$\alpha = 1 - N_{\text{short}}/N_{\text{total}}$	16	7
Mean segment duration	5	0
Segment count	5	0

Table 4: Spearman correlation between structural covariates and the flip rate, at series level and with collections as the unit, plus the range under leave-one-collection-out. Series-level p -values are permutation-based and report the resolution floor of the test. α is the only covariate that does not survive.

Covariate	Series ρ	Coll. ρ	Coll. p	LOCO range	Worst drop
α	0.3241	0.0563	0.827	[0.0898, 0.3741]	TAO
Mean segment duration	0.3717	0.3750	0.138	[0.2408, 0.4814]	GHL
Segment count	-0.3775	-0.4118	0.098	[-0.4252, -0.2535]	TAO
Anomaly density	-0.3601	-0.3333	0.195	[-0.4088, -0.2144]	GHL
Series length	0.2047	0.2696	0.289	[0.0394, 0.3452]	GHL

The short-anomaly ratio is such a covariate. Table 3 shows that α is constant within 16 of 17 collections, and within 7 of the 8 collections that contribute at least eight series. It is, operationally, a collection label. Mean segment duration and segment count are not: both vary within every one of the large collections.

The consequences are in Table 4. At the series level α correlates with the flip rate at $\rho = 0.3241$ with a permutation p at the resolution floor of the test—an apparently decisive result. At the collection level the same association is 0.0563 with $p = 0.827$. Leave-one-collection-out is equally stark: removing TAO alone takes the series-level correlation to 0.0898. The $\alpha = 0$ stratum on this grid consists of eleven TAO series and one other, so the low- α regime is one collection wearing the clothes of twelve observations.

The same holds on the 6-dataset grid, independently. Only 2 of 6 datasets have $\alpha < 1$, so the entire structural claim rests on them; and of the 14 deep-model flips, 6 come from PSM and 5 from SWaT. SWaT has $\alpha = 1.0000$ —the regime in which the structural account predicts the fewest flips.

Finally, spread within the $\alpha = 1$ regime exceeds spread across α regimes: among collections with $\alpha = 1$ throughout, pooled flip rates run from 0.1733 (GECCO) to 0.5876 (GHL), a wider range than the α -stratified contrast itself.

Interval estimates. The same non-independence affects uncertainty. Resampling series gives a 95% interval of [0.2974, 0.3319] for the TSB-AD-M flip rate; a cluster bootstrap over collections gives [0.2701, 0.3695], about three times wider. On the 6-dataset grid the effect is severe enough to be disqualifying: the deep-model interval is [0.1333, 0.3500] at pair level but [0.0667, 0.4167] clustering over the six datasets. A study of that size cannot resolve a flip rate to better than a range spanning most of the unit interval, and should not be used to argue about its magnitude.

6 A worked failure case: interpolation metrics at their endpoints

One response to metric disagreement is to combine metrics. In earlier work we proposed such a composite, which we called SAEScore, with the structure weight α selecting the blend:

$$\text{SAEScore}(D, M) = (1 - \alpha(D)) \text{AUC-ROC}(M) + \alpha(D) \text{Aff-F1}(M), \quad \alpha(D) = 1 - \frac{N_{\text{short}}(D)}{N_{\text{total}}(D)}. \quad (1)$$

We then stratified the AUC-ROC-versus-SAEScore flip rate by α and reported the resulting gradient—0.0000, 0.1758, 0.3321 across $\alpha = 0$, $0 < \alpha < 1$, and $\alpha = 1$ —as confirming a structural regime split.

It confirms nothing. Equation 1 makes SAEScore identically AUC-ROC at $\alpha = 0$ and identically Aff-F1 at $\alpha = 1$; on our artifact the maximum absolute differences are 0.0 and 0.0 respectively. The $\alpha = 0$ stratum therefore compares AUC-ROC with itself and contributes 0 flips out of 3,592 by definition, and the $\alpha = 1$ stratum is bit-identical to the AUC-ROC-versus-Aff-F1 comparison (13,349/40,200 in both). The gradient is linear interpolation between a self-comparison and a fixed comparison. No configuration of the data could have produced anything else.

This generalizes past our own metric. Any composite defined as a convex combination of two metrics, with a weight that reaches its endpoints on real data, will exhibit a monotone “regime” pattern when stratified by that weight, whether or not the underlying structural hypothesis is true. Such a stratification is a description of the metric’s definition, not a finding about benchmarks. The diagnostic is cheap: check whether the extreme strata of the weight reduce the composite to one of its constituents, and if they do, report the constituent comparison instead.

Doing that here gives strata of 0.1466, 0.2897, and 0.3321. These are metric-independent and still monotone—but as Section 5 shows, the gradient is carried by collection identity, and α does not survive as a predictor.

7 What survives

The structural hypothesis is not dead; its standard operationalization is. Mean segment duration correlates with the flip rate at 0.3717 at the series level and 0.3750 at the collection level, and stays within [0.2408, 0.4814] under leave-one-collection-out. Segment count behaves symmetrically at -0.3775 and -0.4118 . Neither collapses the way α does, and neither is constant within the large collections.

We state the limits plainly. With 17 collections the collection-level tests are underpowered: the permutation p -values are 0.138 and 0.098, and we do not claim significance for them. What we claim is the comparison. Under identical treatment, α moves from 0.3241 to 0.0563 and collapses under leave-one-out, whereas duration and segment count retain their sign and roughly their magnitude at both levels. That contrast is the evidence; the individual collection-level coefficients are not.

Note also that these covariates are not independent of one another. Mean segment duration is algebraically determined by anomaly density, series length and segment count, so a regression that conditions on all three cannot say anything further about duration. Reporting several one-at-a-time associations, as in Table 4, is the honest presentation at this sample size.

8 A reporting protocol

The following costs no additional experiments and would have caught every issue above.

1. Report the null. State the random-ranking baseline next to any metric-disagreement statistic. For pairwise flip rates it is 0.5; obtain it by permuting one metric across models within each evaluation group.
2. Stratify by margin. Report the flip rate as a function of the primary metric’s gap, or at minimum report it with near-ties excluded and say what fraction of pairs that removes.
3. Cluster at the source. Use the source collection or dataset, not the series, as the resampling unit, and report the clustered interval. Where the two intervals differ materially, report both.
4. Check identifiability before conditioning. Before explaining disagreement with a covariate, check whether that covariate varies within clusters. If it does not, a series-level association is a between-collection association with an inflated n .
5. Leave one collection out. Report the range of any structural association under removal of each collection in turn. A result carried by one collection should be presented as such.

6. Do not stratify a composite by its own weight. If a metric reduces to a constituent at the extremes of a weight, stratifying by that weight measures the definition.

9 Related work

Our object of study is a statistic, not a metric, so the closest work is methodological rather than metric-proposing.

Critiques of TSAD evaluation. Kim et al. (2022) showed that point-adjustment lets random scores outperform published detectors, which is the canonical demonstration that a TSAD metric can be uninformative. Wu and Keogh (2023) and Liu and Paparrizos (2024) argue that benchmark datasets themselves are flawed, mislabeled, or trivially solvable, and Schmidl et al. (2022) and Boniol et al. (2024) survey the resulting landscape. Lyu (2026) stress-tests post-point-adjustment replacement metrics and finds that several, including affiliation-based and ROC-based scores, inflate steeply under best-of- N seed selection while PR-based metrics stay flat. That result is complementary to ours and sharpens it: the metric on one side of our comparison has a known instability under selection, which is a further reason not to read a raw flip count as a property of the benchmark.

Replacement metrics. Range-based precision and recall (Tatbul et al., 2018), affiliation metrics (Huet et al., 2022), VUS (Paparrizos et al., 2022), PATE (Ghorbani et al., 2024), DQE (Li et al., 2026), and confidence-consistency evaluation (Zhong et al., 2025) each repair a specific deficiency of point-wise scoring. Wagner et al. (2025) approaches the space axiomatically, specifying properties a metric ought to satisfy, and Yang et al. (2025) organize the resulting metric space by the evaluation problem each metric targets—including a dimension for robustness against random-score inflation, which is chance baselining applied to metric values. Our contribution is orthogonal to all of these: whichever metric one adopts, comparisons between metrics are currently reported without the ingredients above.

Ranking instability as a statistic. Quantifying how far rankings move under a change of protocol is now common practice across ML evaluation, from preference-data perturbations in language model leaderboards (Huang et al., 2025) to protocol axes in computational biology (Kendiukhov, 2026). The latter is the closest methodological neighbour to our Section 3, and reaches the same conclusion in a different domain. It does not address clustering; its own limitations section lists confounding with tissue identity as unresolved, which is the analogue of the problem we take up in Section 5.

Statistical practice in benchmark evaluation. The concern that evaluation units are not independent replicates is old outside machine learning (Hurlbert, 1984). Within it, the argument that point estimates on small benchmark suites mislead without interval estimates has been made forcefully in reinforcement learning (Agarwal et al., 2021) and for language model evaluations (Miller, 2024), and Röchner et al. (2025) argue on general grounds that anomaly detection benchmarks should be organized into scenario groups sharing relevant characteristics. Ours is the quantitative counterpart to that last argument: grouping is not only a taxonomic convenience but the difference between a structural association of 0.3241 and one of 0.0563. Relative to Agarwal et al. (2021) and Miller (2024), which resample runs or items as exchangeable units, our point is that the exchangeable unit here is the source collection, and that identifying it changes which covariates remain admissible at all.

10 Limitations

Our conclusions are conditioned on a specific metric pair, AUC-ROC versus Aff-F1 at cardinality one with contamination-rate thresholding. Other pairs may behave differently, and the thresholding rule is known to matter (Masakuna et al., 2024). The TSB-AD-M slice follows file-ID order and is model-coverage-limited rather than a complete benchmark run, so it should be read as a replication of the

statistical phenomenon and not as a TSB-AD leaderboard claim. With 17 collections and 6 datasets, the cluster-level analyses that carry our main argument are themselves underpowered—which is part of the point, but does mean our positive claims about duration and segment count are directional rather than established. We have not shown that any published conclusion in the TSAD literature is wrong as a result of these omissions; we have shown that the evidence offered for one class of conclusion, including ours, does not have the strength attributed to it.

11 Conclusion

A rank-flip rate is a compact and persuasive statistic, and it is currently reported in a way that cannot be interpreted. Supplied with a null, the flip rates in this literature and in our own prior work describe metrics that substantially agree. Stratified by margin, a real residual disagreement remains that near-ties do not explain. Clustered at the source collection, the structural account of that disagreement does not survive, and neither does the composite metric we built on it. What we can defend is narrower than what we previously claimed, and we think it is more useful: point and segment metrics disagree on roughly one in five clearly separated model pairs, that disagreement is heterogeneous across benchmark collections to a degree that dwarfs any anomaly-structure effect we can identify, and the structural covariates that do track it are continuous ones the field has not been using.

Broader Impact Statement

This work audits an evaluation statistic and proposes reporting practice; it introduces no model, no dataset, and no deployable artifact. The direct effect we anticipate is that some published claims about metric instability in time-series anomaly detection will be restated more conservatively, including our own. We see one indirect risk worth naming. Showing that a widely cited instability figure overstates disagreement could be read as licence to keep reporting single point-wise metrics on benchmarks whose anomalies are sustained segments. That reading would be wrong: our margin-stratified result shows that substantive disagreement persists among clearly separated detectors, and the labeling mismatch documented in prior work is unaffected by anything here. The benchmarks we use are public industrial and infrastructure datasets containing no personal data.

Reproducibility statement

All results are recomputed from committed JSON artifacts by three scripts, and every numeric claim in this manuscript is generated into numbers.tex by a fourth, so the text cannot diverge from the analysis. The statistical helpers are implemented without a scipy dependency and are unit-tested against hand-computed values. The repository includes the derived metric rows, the taxonomy summary, and the reproduction commands; it does not redistribute the source benchmark datasets, for which access instructions are provided.

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